

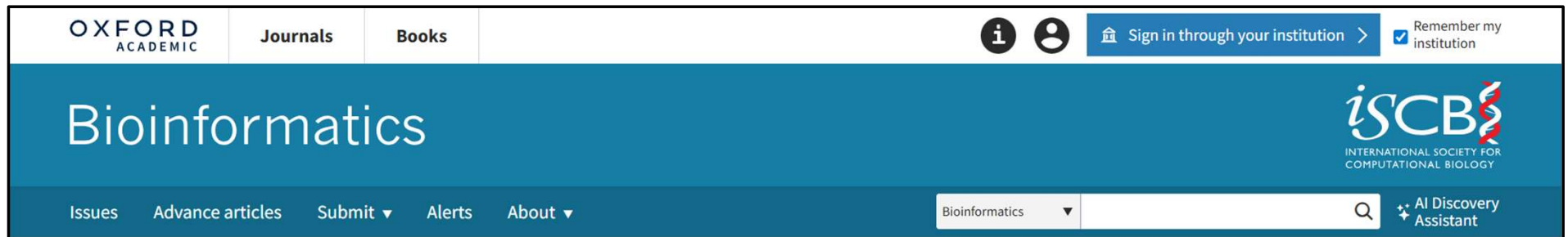
Explainable Multilayer Graph Neural Network for cancer

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Paper



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Problem

EMOGI rely on a single PPI network, which means that cancer gene prediction results can vary greatly depending on the network chosen. This reflects the limitation that a single network cannot fully capture the complex molecular interactions underlying cancer.

GNN (message-passing scheme)

$$\mathbf{m}_u^{(l)} = \text{Aggregate}^{(l)} \left(\{ \mathbf{h}_v^{(l-1)} : v \in \mathcal{N}(u) \} \right), \quad (1)$$

$$\mathbf{h}_u^{(l)} = \text{Combine}^{(l)} \left(\mathbf{h}_u^{(l-1)}, \mathbf{m}_u^{(l)} \right), \quad (2)$$

$\mathbf{m}_u^{(l)}$: aggregated message

$\mathbf{h}_u^{(l)}$: the hidden representation of node u at the l^{th} layer of the GNN architecture

GCN : using fixed weights

$$h'_u = W^\top \sum_{v \in \mathcal{N}(u) \cup \{u\}} \frac{h_v}{\sqrt{\hat{d}_v \hat{d}_u}}$$

GAT : using learnable attention scores

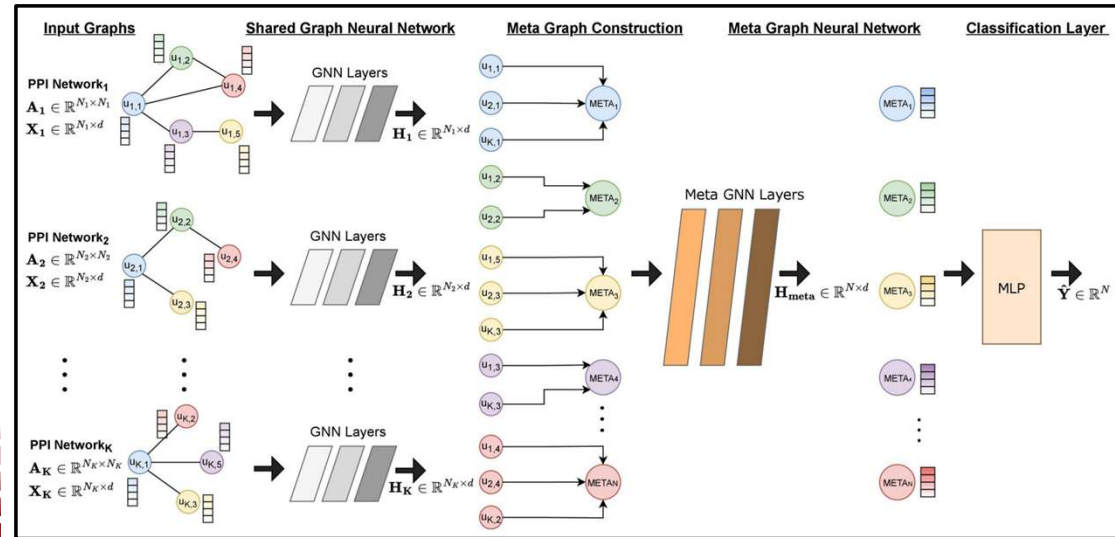
$$\alpha_{u,v} = \frac{\exp \left(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_u \parallel \mathbf{W}\mathbf{h}_v]) \right)}{\sum_{k \in \mathcal{N}(u) \cup \{u\}} \exp \left(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_u \parallel \mathbf{W}\mathbf{h}_k]) \right)}.$$

Data

Omics data: TCGA (Pan-cancer data)

Network data: CPDB, STRING-db, Iref, Iref(newest version), Multinet, and PCNet

EMGNN architecture



$$H^{(i)} = f_1(X^{(i)}, A^{(i)})$$

$$H_{\text{meta},j} = f_2(X_{\text{meta},j}, A_{\text{meta},j})$$

$$\hat{y} = f_3(H_{\text{meta},j})$$

$$\hat{Y} = f_3(f_2(f_1(X, A), A_{\text{meta}}))$$

Model interpretation

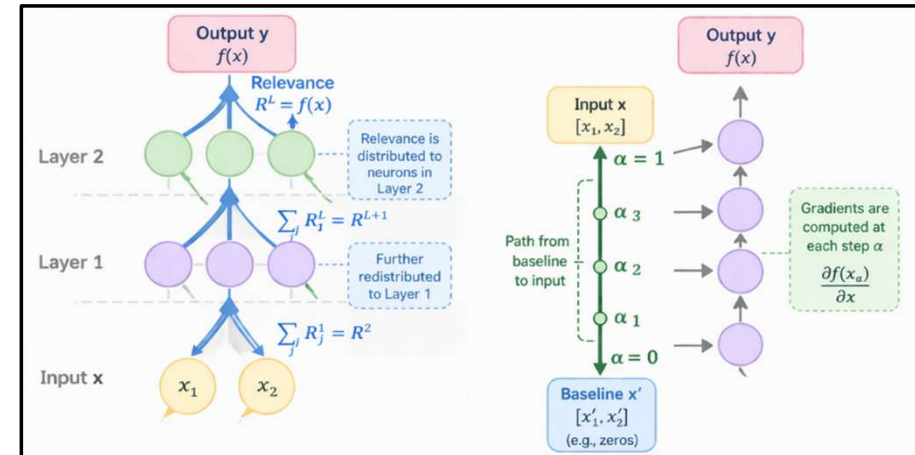
IG (Integrated Gradients , EMGNN model)

Method that computes the integral of gradients along the path from a reference baseline to the actual input

LRP (EMOGI model, Layer-wise Relevance Propagation)

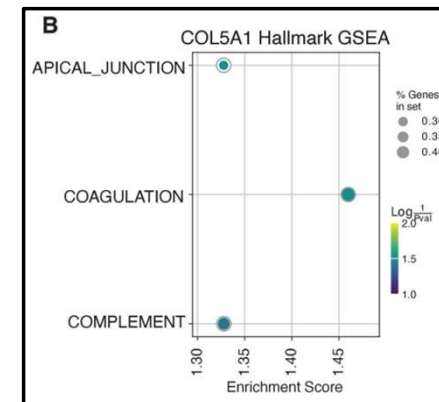
A method that propagates the model output backward through the network layers to reassign contribution scores (**Relevance scores**) to individual neurons/features

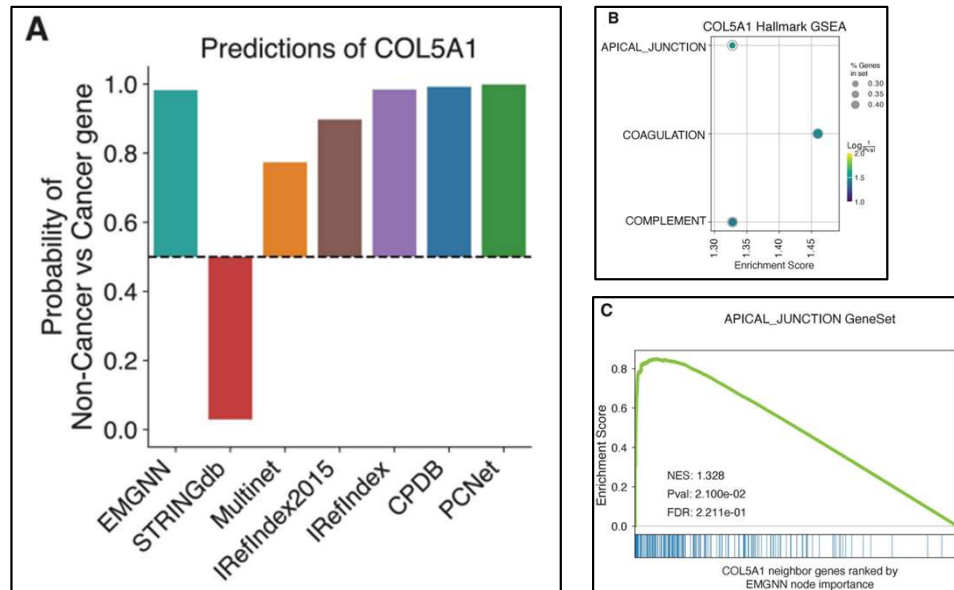
$$\text{IntegratedGrads}_i(x) = ((x_i - \hat{x}_i) \int_{\alpha=0}^1 \frac{\partial F(\hat{x} + \alpha(x - \hat{x}))}{\partial x_i} d\alpha$$



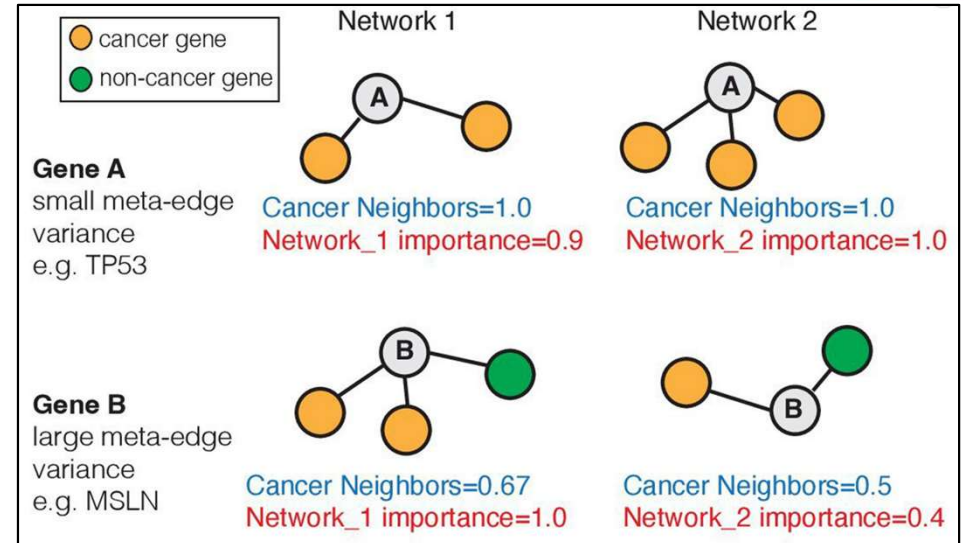
GSEA (Gene Set Enrichment Analysis)

1. Calculate the importance of **neighboring nodes** using the **IG method**.
2. Exclude genes with zero contribution and **rank** the remaining genes.
3. Compare with **cancer hallmark gene sets**.
4. Assess whether genes belonging to **specific biological pathways** are enriched at the top of the ranking, and calculate the P-value and FDR (false discovery rate).





Cancer neighbors

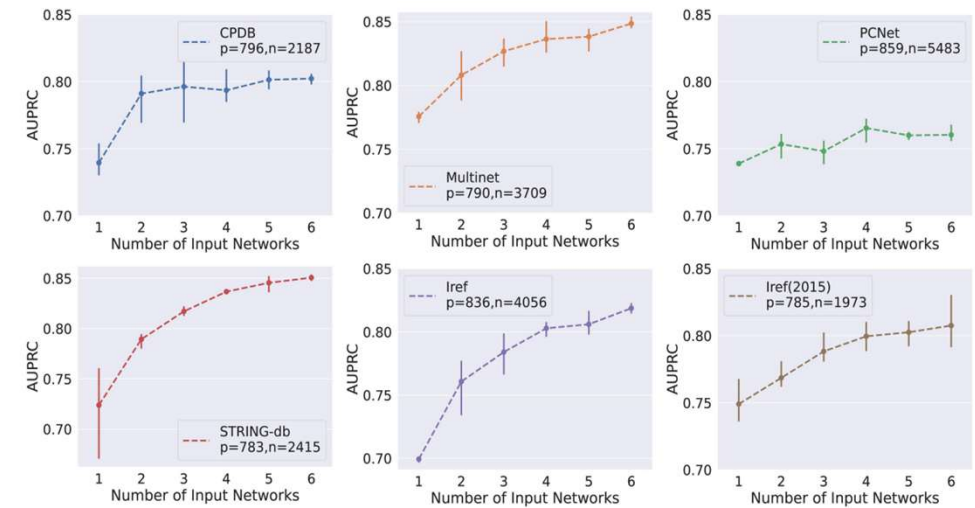


Performance of EMGNN

Table 1. Test AUPRC values and standard deviation across different PPI networks across five different runs.

Method	CPDB	Multinet	PCNet	STRING-db	Iref	Iref(2015)
Random	0.27	0.18	0.14	0.24	0.17	0.28
20/20+	0.66	0.62	0.55	0.67	0.61	0.65
MutSigCV	0.38	0.33	0.27	0.41	0.35	0.43
HotNet2 diffusion	0.62	0.56	0.48	0.50	0.45	0.65
DeepWalk+features RF	0.74	0.71	0.72	0.71	0.66	0.71
PageRank	0.59	0.53	0.54	0.44	0.42	0.62
GCN without omics	0.57	0.53	0.47	0.39	0.37	0.64
DeepWalk + SVM	0.73	0.51	0.63	0.52	0.62	0.66
RF	0.60	0.59	0.51	0.61	0.54	0.62
MLP	0.58	0.63	0.47	0.63	0.55	0.64
EMOGI(Schulte-Sasse <i>et al.</i> 2021)	0.74	0.74	0.68	0.76	0.67	0.75
EMOGI(Majority Vote)	0.768	0.783	0.673	0.781	0.676	0.775
EMOGI(Hong <i>et al.</i> 2022)	0.775 ± 0.003	0.732 ± 0.003	0.745 ± 0.002	0.763 ± 0.003	0.701 ± 0.004	0.757 ± 0.001
EMGNN(GCN)	0.809 ± 0.006	0.854 ± 0.007	0.761 ± 0.001	0.856 ± 0.002	0.822 ± 0.002	0.800 ± 0.010
EMGNN(GAT)	0.776 ± 0.018	0.796 ± 0.034	0.730 ± 0.031	0.805 ± 0.030	0.739 ± 0.033	0.773 ± 0.049

Bold numbers indicate the best results.

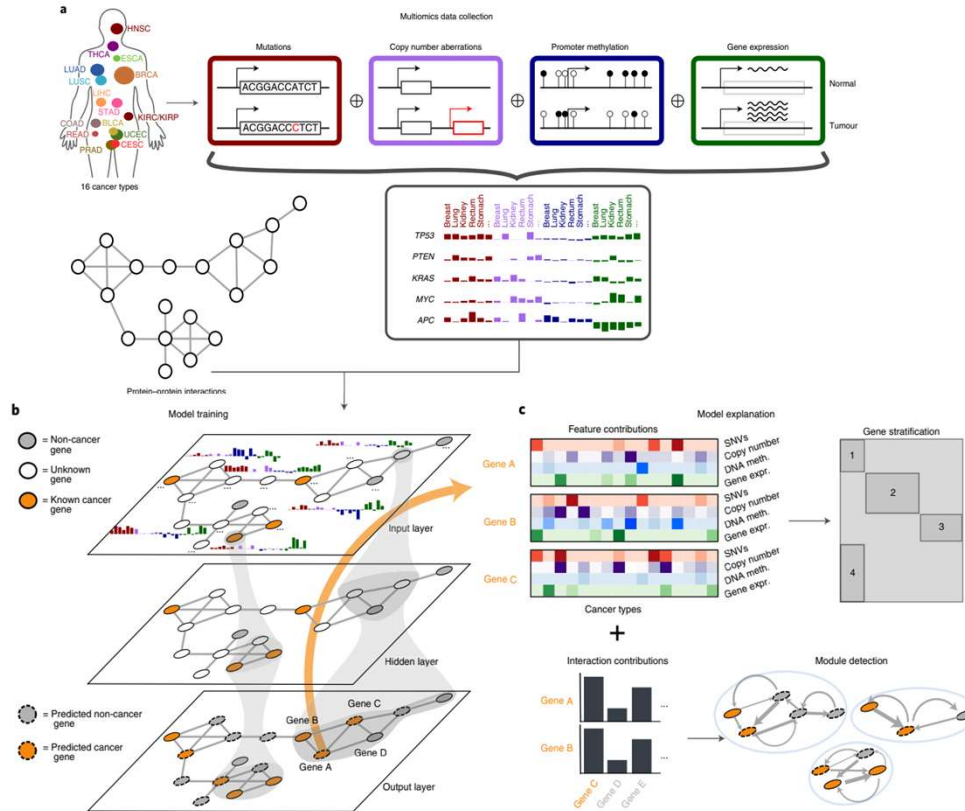


Take-home Messages

- Go beyond the limitations of a single network by modeling multi-layered topological relationships. (Meta node, Meta graph)
- Use the IG method to analyze specific causal genes, and employ GSEA to explain why these genes are associated with cancer.
- Present a vision for extending beyond homogeneous graphs to tissue-specific networks or heterogeneous graphs in the future.

Thank you

EMOGI architecture

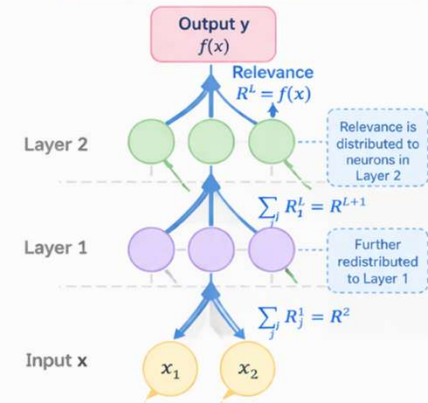


LRP vs IG

LRP (Layer-wise Relevance Propagation)

Redistributes the output relevance backward to lower layers while preserving the total relevance.

Key Idea: Backward redistribution with "conservation principle"



Conservation Principle

$$\sum \text{all inputs } R_i = \text{Output } f(x)$$

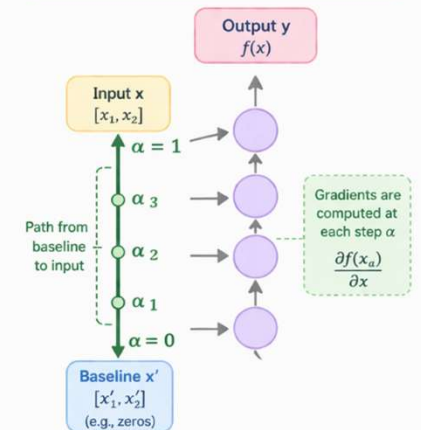
LRP Rule (example)

$$R_i^{(l)} = \sum_j \frac{x_i w_{ij}}{\sum_i x_i w_{ij} + \epsilon} R_j^{(l+1)}$$

IG (Integrated Gradients)

Computes the integral of gradients from a baseline to the input, attributing contribution along the straight path.

Key Idea: Path integral of gradients from baseline to input



Integrated Gradients Formula

$$IG_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i} d\alpha$$

Completeness (Conservation)

$$\sum_i IG_i(x) = f(x) - f(x')$$

Total attribution equals the output difference.

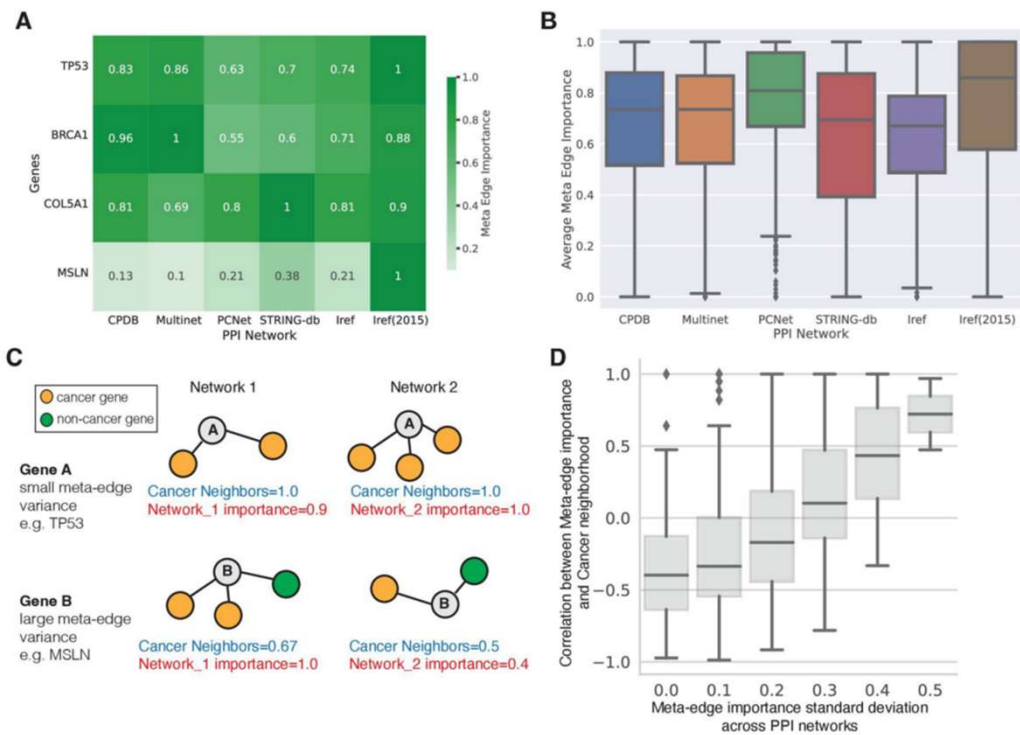


Figure 3. Explanation of each PPI network's contribution to cancer gene predictions. (A) Representative PPI network contributions in known cancer genes and newly predicted cancer genes. TP53 and BRCA1 are known cancer genes; COL5A1 and MSLN are newly predicted cancer genes. (B) Overall distribution of meta-edge feature importance for all known cancer genes across six PPI networks. Meta-edge feature importance was normalized to one (see Section 2 for details). (C) A hypothetical illustration of PPI network cancer neighborhood implicated in the variation of meta-edge importance. (D) Empirical analysis demonstrates a higher correlation between meta-edge importance and cancer neighborhood for genes with a large meta-edge variance.

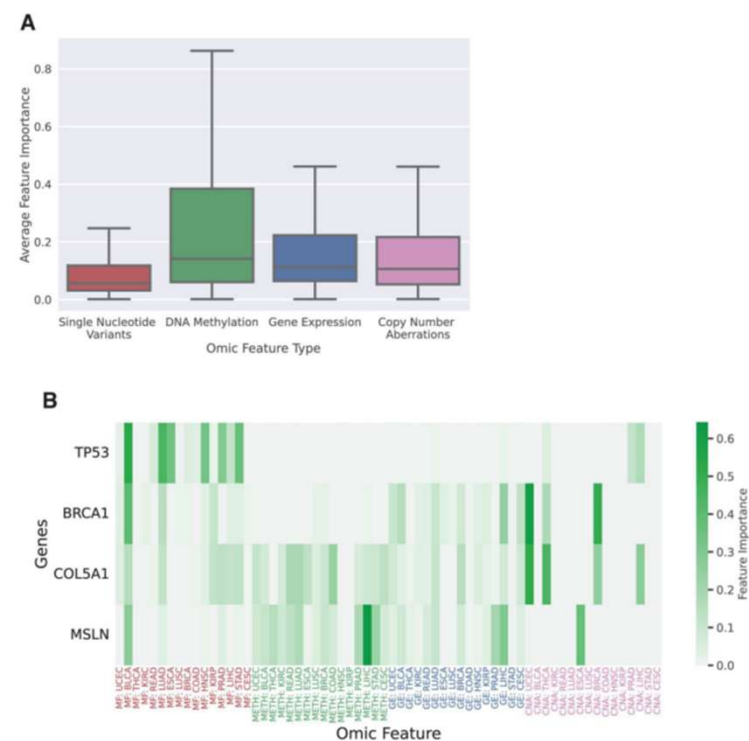


Figure 4. Explanations of multi-omic node feature importance in cancer gene predictions. (A) Overall distribution of node feature importance grouped by omic feature types, including single-nucleotide variants (SNV), DNA methylation (METH), gene expression (GE), and copy number aberrations (CNA), for known cancer genes. (B) Detailed node feature importance for the four genes analysed in Fig. 3B. X-axis labels were color-coded to match the omic feature types in (A). Individual tumor types were coded according to TCGA study abbreviations (Weinstein et al. 2013).

LRP의 원리

LRP가 필요한 이유

딥러닝 모델은 보통 Black box임 (모델이 예측은 가능하지만 그 이유를 알기 어렵다. 그래서 사용하는 것이 LRP)

LRP의 핵심 아이디어

LRP는 모델의 예측값을 입력 feature 로 다시 분배하는 방법이다. 즉 prediction score -> layer 별로 relevance 분배 -> input feature 까지 전달 -> 어떤 input이 prediction에 얼마나 기여했는지 알 수 있다.

직관적인 원리

Ex) 모델이 Fluorescence =0.9 라는 예측을 했다고 하자.

LRP는 이 0.9라는 값을 거꾸로 내려보내면서 neuron -> neuron -> input 에 각 feature 기여도를 나눈다.

결과 : '각 위치의 중요도 ' 가 나온다. (결과는 보통 heatmap으로 나타냄)

A position12 -> +0.3

G position 15 -> +0.25

T position 9 -> +0.1

수식개념

LRP의 기본규칙은 relevance conservation. 즉, 상위 layer relevance 합 = 하위 layer relevance 합

수식 형태:

$$R_j = \sum_k \frac{a_j w_{jk}}{\sum_j a_j w_{jk}} R_k$$

의미

기호

a_j

w_{jk}

R_k

R_j

의미

neuron activation

weight

상위 layer relevance

하위 layer relevance

즉 activation × weight 비율로 relevance를 분배한다.